

Intelligent Document Extraction & Validation Platform

AI Engineer Internship Technical Case Study Presentation | SentinelAI

- **Candidate:** Partha Protim Mondal
- **Core Architecture:** FastAPI, PyMuPDF, Multimodal LLMs (Gemini / OpenAI), SQLite, Bootstrap 5 UI
- **Documents in Scope:** Invoices, Balance Sheets, Profit & Loss Statements, Cash Flow Statements
- **Mandatory Deliverables:** Public GitHub Repository, Live Deployed Platform, Swagger API (/docs), 20/20 Automated Tests

1. Problem Statement & Core Objectives

Addressing manual bottlenecks in financial services document processing

- **The Challenge:** Financial services teams receive native and scanned PDF, JPG, and PNG documents with diverse layouts, noise, and variable line items.
- **Comprehensive Ingestion:** Supports 4 document types with automated input integrity checks (page limit ≤ 3 , file type validation).
- **Accurate Field & Table Extraction:** Extracts key header values, line items, and grounding evidence snippets without hallucinating missing values.
- **Automated Financial Reconciliation:** Performs domain math checks (Balance Sheet equality, P&L; profit reconciliation, Cash Flow sum, Invoice total).
- **Full Deployment:** Ready for evaluation via live public URLs (Render/Railway) and interactive web dashboard.

2. End-to-End System Architecture

Modular microservice design ensuring clean separation of concerns

- **FastAPI Application Gateway:** Asynchronous REST API exposing `/api/v1/documents/process`, `/api/v1/documents/{name}`, and `/api/v1/health`.
- **Input Validation Service:** Enforces MIME type checking, corruption detection, and page limit (≤ 3 pages).
- **OCR & Multimodal Vision Engine:** PyMuPDF rasterization for high-resolution page images + multimodal LLM extraction adapter.
- **Financial Validation Engine:** Rules engine calculating mathematical reconciliations, variances, and status tagging (PASS / FAIL / NOT_APPLICABLE).
- **Persistence & Frontend:** SQLite ORM with repository pattern, paired with responsive Bootstrap 5 web dashboard.

3. Financial Validation Engine & Formulas

Rigorous mathematical reconciliation across all four financial statements

- **Invoices:** $\text{Quantity} * \text{Unit Price} \approx \text{Line Total}$; $\text{Sum}(\text{Line Totals}) \approx \text{Subtotal}$; $\text{Subtotal} + \text{Tax} - \text{Discount} \approx \text{Total Amount}$.
- **Balance Sheet:** $\text{Total Capital \& Liabilities} \approx \text{Total Assets}$ (evaluated independently for current and comparative prior years).
- **Profit & Loss:** $\text{Total Income} \approx \text{Interest Earned} + \text{Other Income}$; $\text{Total Expenditure} \approx \text{Expended} + \text{Operating} + \text{Provisions}$; $\text{Net Profit} \approx \text{Income} - \text{Expenditure}$.
- **Cash Flow Statement:** $\text{Operating} + \text{Investing} + \text{Financing} + \text{FX} \approx \text{Net Change in Cash}$; $\text{Opening Cash} + \text{Net Change} \approx \text{Closing Cash}$.
- **Accounting Brackets:** Automatically converts negative financial formatting like (1,018,904,990) to -1018904990.0.

4. REST API & Web Dashboard Interface

Intuitive user experience paired with robust backend standards

- **POST /api/v1/documents/process:** Synchronous multipart upload endpoint returning Section 5.2 compliant JSON response.
- **GET /api/v1/documents/{document_name}:** Retrieves latest processed results by document name, with 404 error handling.
- **GET /api/v1/documents:** Lists all processed records for dashboard table population.
- **GET /api/v1/health:** Liveness probe endpoint returning healthy status.
- **Interactive Dashboard:** Drag-and-drop file upload, real-time spinner, validation badges, and 1-click Raw JSON modal viewer.

5. Automated Testing & Verification

20/20 automated tests passing across validation, extraction, and APIs

- **Input Validation Tests (test_validation.py):** Valid PDFs, images, rejection of unsupported types, 0-byte files, and > 3 page limits.
- **Domain Math Tests (test_extraction.py):** Negative bracket parser, all 4 document reconciliation checks, intentional variance detection.
- **API Integration Tests (test_api.py):** Health check, multipart process flow, document retrieval, and dashboard list endpoints.
- **Missing Field Handling:** Verifies that missing fields return NOT_APPLICABLE status instead of guessing or inventing values.

6. Production Scalability & Deployment Strategy

Architected for seamless migration to enterprise cloud infrastructure

- **Containerization:** Production Dockerfile with system dependencies, healthcheck, and multi-stage optimizations.
- **Cloud Deployment:** 1-click blueprint for Render, Railway, or Koyeb via render.yaml and Procfile.
- **Async Queueing:** Roadmap includes Celery + Redis task queue for background processing of large document batches.
- **Human-in-the-Loop (HITL):** Low-confidence extractions (< 85%) or validation variances automatically flagged for manual review.
- **Security:** Zero secrets committed to source control; environment variable injection; secure file handling.